Darren Ke

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EDUCATION

Yale University, New Haven, CT — Expected Graduation May 2028

- Bachelor of Science, Mechanical Engineering (ABET-Accredited), Energy Studies Certificate, 3.67/4.0 GPA
- *Activities/Distinctions*: Yale American Society of Mechanical Engineering, Yale American Institute of Aeronautics and Astronautics, Science Technology and Research Scholars Fellowship, Yale Fishing

ENGINEERING EXPERIENCE

Pneumatic Pellet Launcher, *Mechanical Design & Systems Development*, Yale Nanoprobe Lab — Jun. 2026 - Present

- Lead development and continuous improvement of a pneumatic high-speed impact tester using an integrated system of compressed air, actuated ball valve, and solenoid capable of firing pellets at speeds of around 100m/s
- Designed and manufactured a glass sample fixture using CAD, water-jetting, and milling to reduce energy interference in high-speed impacts, improving reliability of tests by around 5%
- Develop hardware and LabVIEW controls to automate sample-stage positioning across multiple samples and holders

Autonomous Rover, *Engineer and Treasurer* — Jan. 2025 - Present

- Lead a 30-member robotic team to develop a competition-ready rover by securing a $15,000 budget, managing funds, reviewing Bill of Materials, and setting project milestones
- Identified fault areas and reduced peak stress concentrations at specific points in the rover wheel using FEA-driven optimization
- Conduct rapid fabrication and prototype iteration with 3D printing and CNC milling to support continuous design validation and testing

Foot-Actuated Kitchen Knife, *Independent Project* — Sep. 2025 - Jan. 2026

- Developed a foot-actuated cutting device using a four-bar linkage, pulley system, and elastic return mechanism, achieving an estimated 100N output force through iterative CAD design, fabrication, and testing

RESEARCH EXPERIENCE

Yale Nanoprobe Lab, *Lead Undergraduate Researcher* — Jun. 2026 - Present

- Lead a Corning Inc.-funded project to study crack morphology in oxide glasses under dynamic high-speed loading
- Measure coefficients of restitution across three compositions of oxide glasses at multiple impact velocities to quantify relationships between energy dissipation and glass composition
- Employ GIXRD, high speed imaging, and microscopy to characterize coatings, residual stresses, and surface conditions and correlate them with fracture responses to high-speed impacts

Gentner Lab, *Lab Assistant* — Sep. 2025 - Jan. 2026

- Developed a data acquisition system to sample data in 150+ homes using an integrated system of Raspberry Pi, Python, and air quality sensors, increasing data storage capabilities and cutting potential monthly costs by more than $14
- Designed and fabricated protective sensor enclosures and compact support structures using CAD and additive manufacturing to streamline package setup, improve weather resistance, and cut potential fabrication costs by $10/unit
- Led a study correlating the effect of QuantAQ orientation on PM10 and PM2.5 measurements using MATLAB to improve the experimental reliability of tests

LEADERSHIP

Pierson College Administrative Office, *Office Aide* — Jan. 2025 - Present

- Serve as a primary point of contact for students and staff to resolve day-to-day issues and coordinate with university administrators on operational and logistical needs
- Lead execution of major college events such as move-in and commencement while managing competing priorities in a fast-paced environment

TECHNICAL SKILLS

- *Digital Design and Fabrication*: Onshape, SolidWorks, Raspberry Pi
- *Programming/Software*: Python, C, C++, MATLAB, Excel, LabVIEW
- *Manufacturing/Machining*: Lathe, CNC, Additive manufacturing, Waterjet, Laser Cutting, Hand tools